ITEC624 - Network Security & Applications

Semester 2 - 2025

Assessment 3

Skill-Based Tasks & Case Study

Submitted By
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Word Count: 2229

Feed Forward:

Reflection Questions	Feedback and Improvement Actions
1. In your opinion, how do you think previous project inputs contributed to the development of this report?	As stated in previous reviews, this needs a more thorough explanation of the technical decisions taken, which I have tried to cover in this report: for example, the considerations made against the costs, performance, and the meeting of objectives set by ABC Company in choosing Azure PaaS and parts of the hardware.
2. For future projects, what type of feedback would you find most helpful?	Detailed feedback is what helps me tune-especially those areas that need work, like technical detail, clarity, or structure. When I know exactly where I need to add evidence or revise my arguments, I can construct stronger, more targeted arguments in future assignments.
3. Were you having problems trying to understand the use of comments from previous assignments?	Some of the previous critiques were too general to locate specific areas that needed attention. For instance, comments that highlight specific passages or ideas that require more careful consideration or evidence are of more value than the general advice to "improve the analysis."
4. How has your strategy changed considering earlier criticism?	Previous remarks emphasized that comments should be short and fact-based. Therefore, in this research work, I tried to give the shortest explanations and examples that will ensure that things are clear and support my recommendations effectively, such as comprehensive estimates of costs for cloud services.
5. About anything, how do you think the feedback process could be improved to help you learn more?	This would also be assisted if recommendations were added concerning readings or resources on hard subjects, such as cloud architecture models. Additionally, apart from this, feedback should include a summary of strengths and areas of development. That would really help me understand and lift the standard of my work in general.

Checklist:

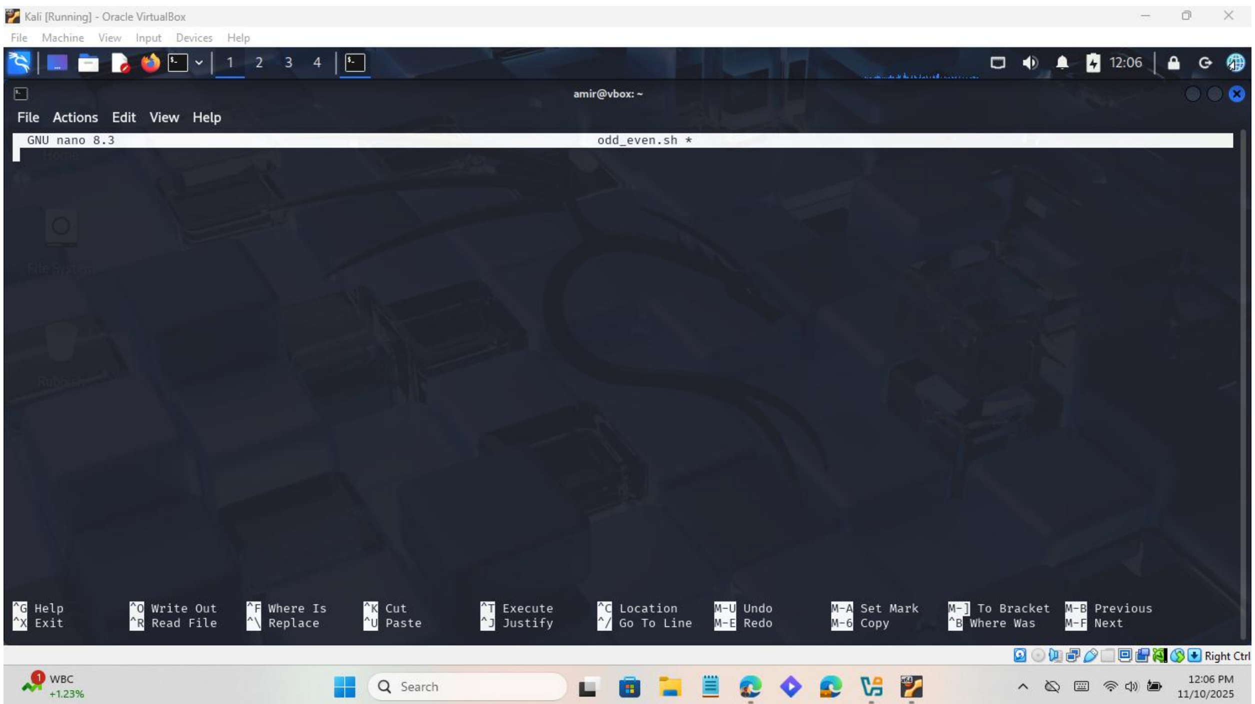
My submitted assignment report is within the specified word limit.	✓
I have included references using the specified referencing style.	✓
I have correctly cited all my sources and references.	✓
I have formatted my report as per the specifications.	✓
I have checked my Turnitin report to ensure the similarity report is within the acceptable level(below 20% similarity)	✓
I have actioned the feedback advice provided to me from the previous assignment feedback.	✓
I have completed proofreading and checked for spelling and grammar.	✓
I have submitted my work before the due date/time.	✓
I have submitted feed feed-forward template along with my assignment submission.	✓

Table of Contents

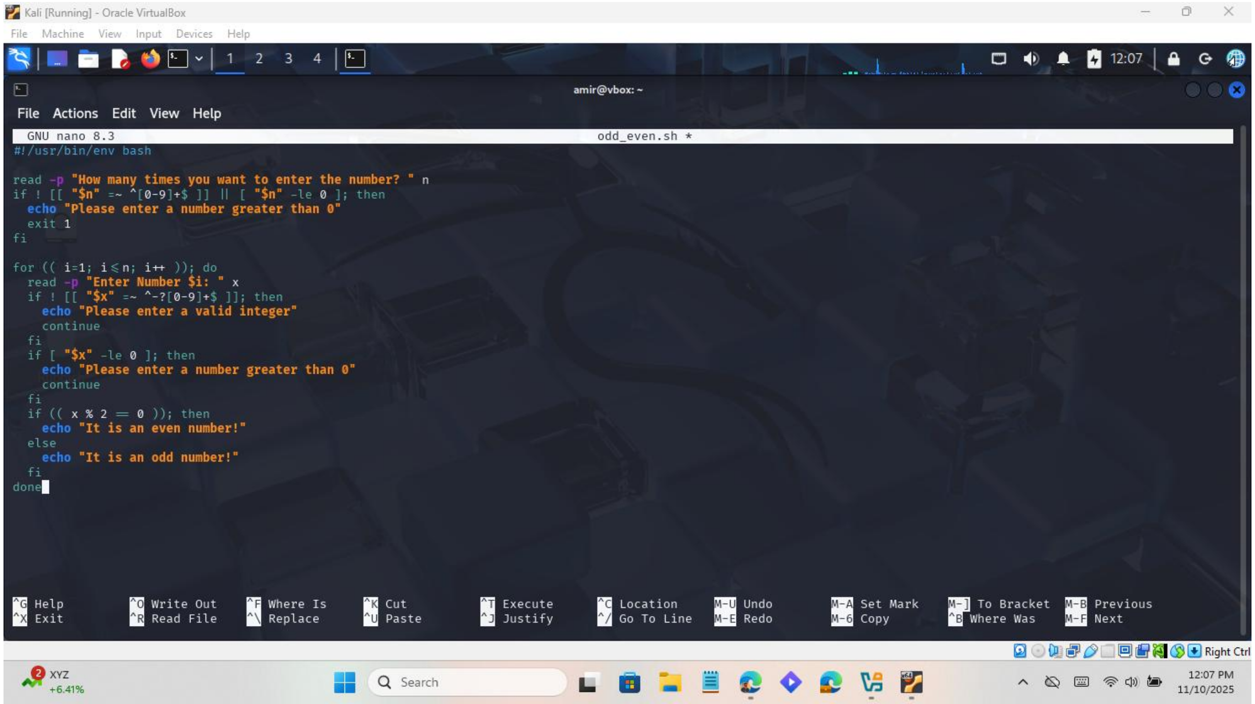
Part 1: Skill-based assessment

Q1. Shell script: odd/even with n repetitions

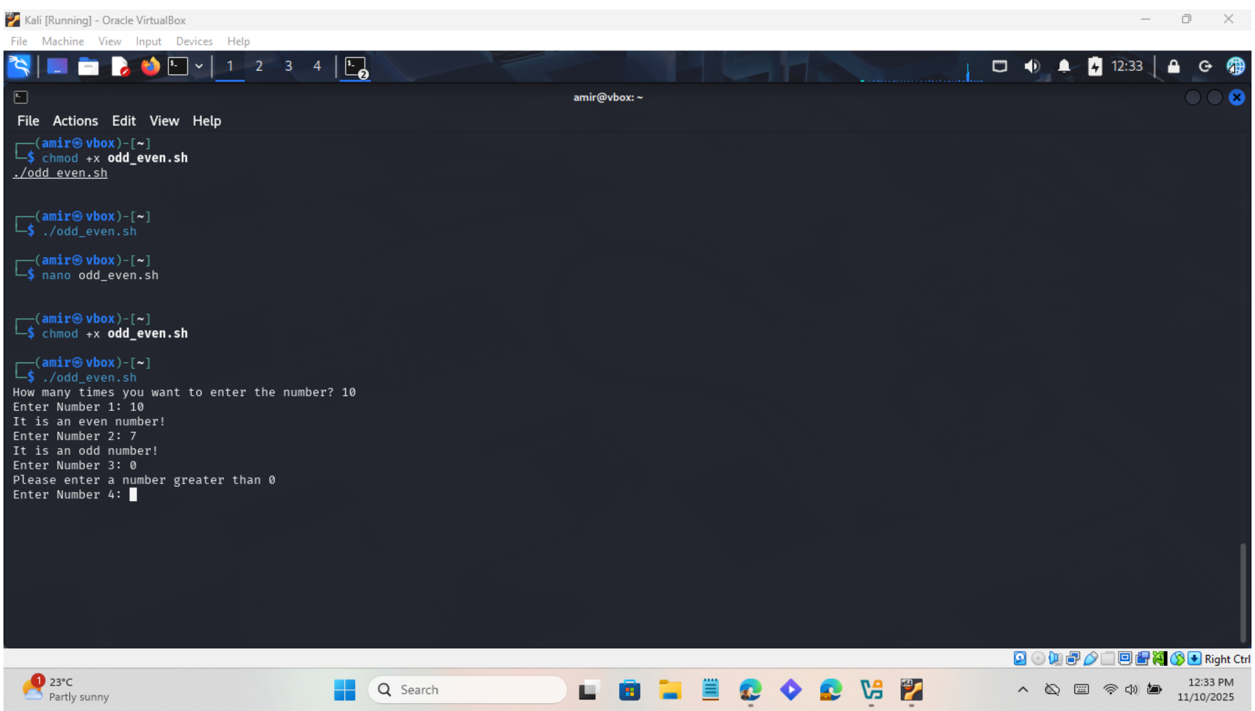
1.1. Create the script (odd_even.sh)



1.2. Code for odd_even.sh file:



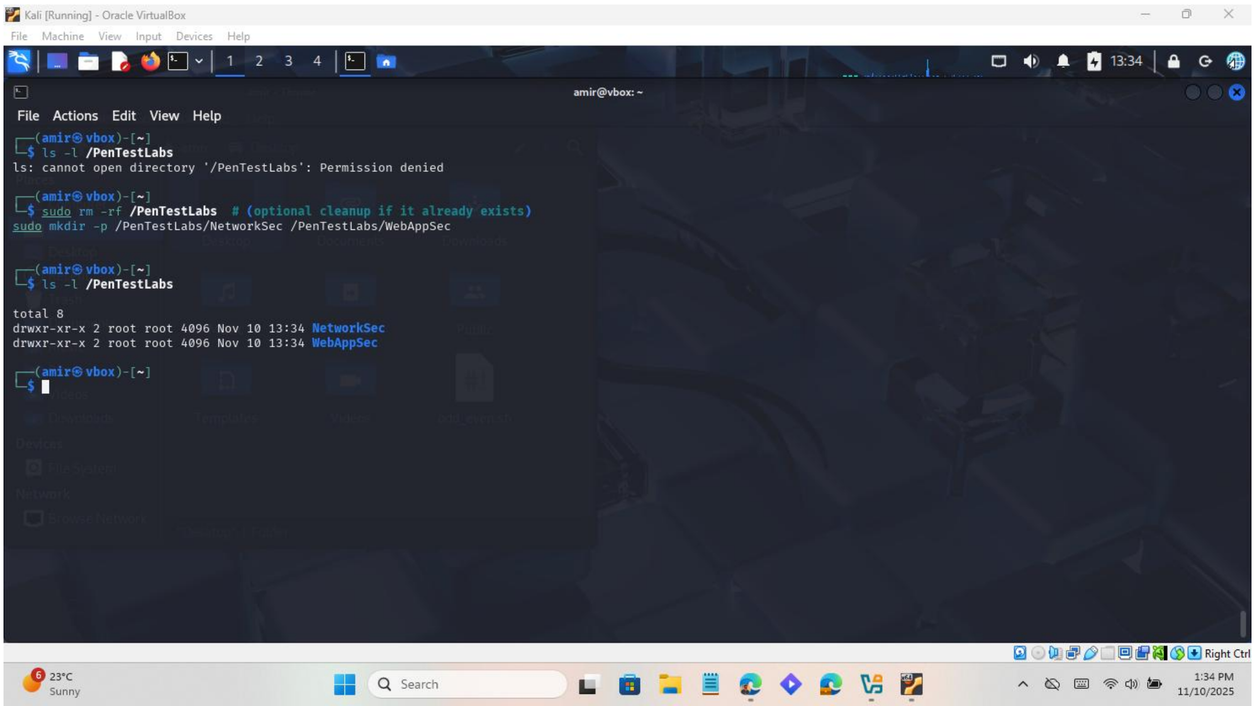
1.3. Make executable and run



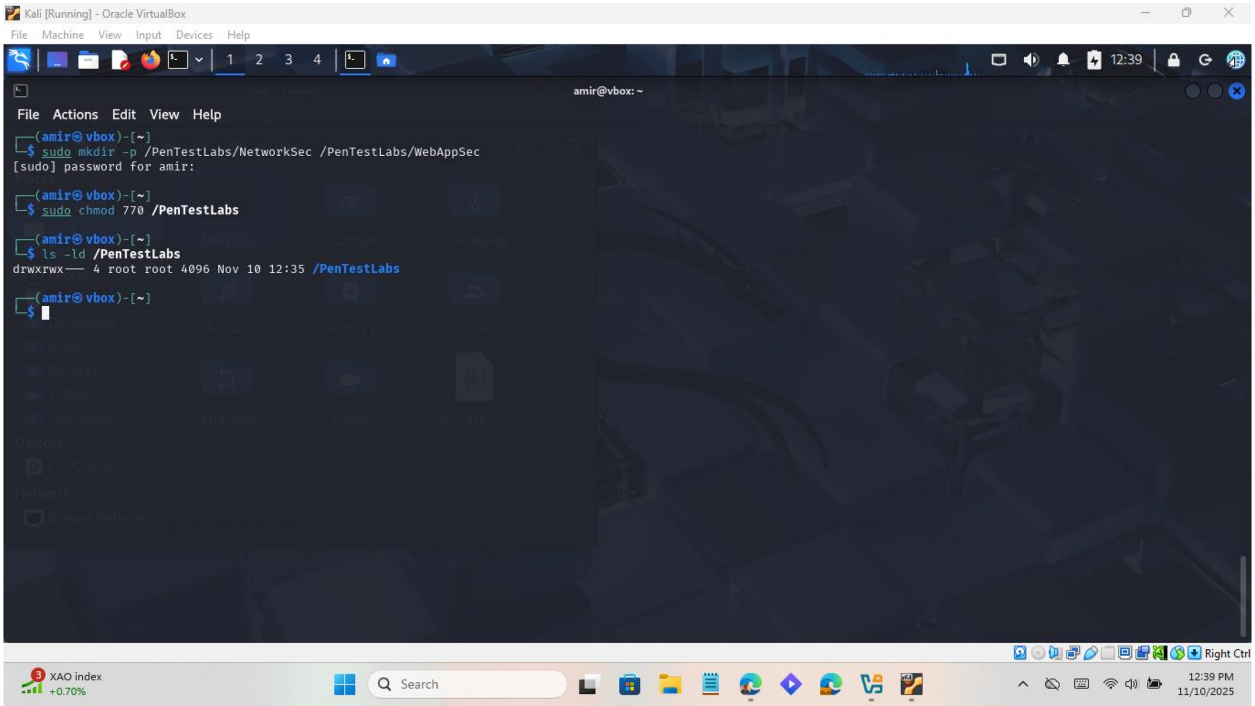
Q2. Advanced directory permissions and ACLs

I will create users/groups, set directory ownership/permissions, and verify with ACLs.

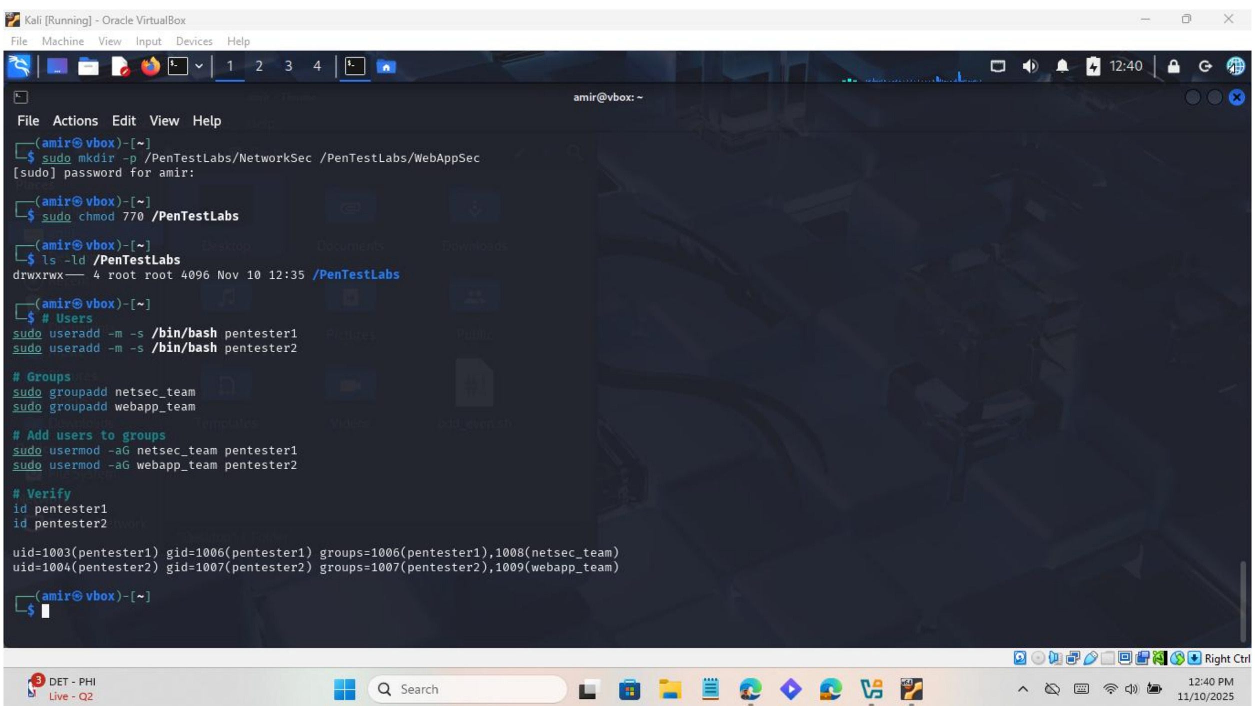
2.1. Create the directory structure



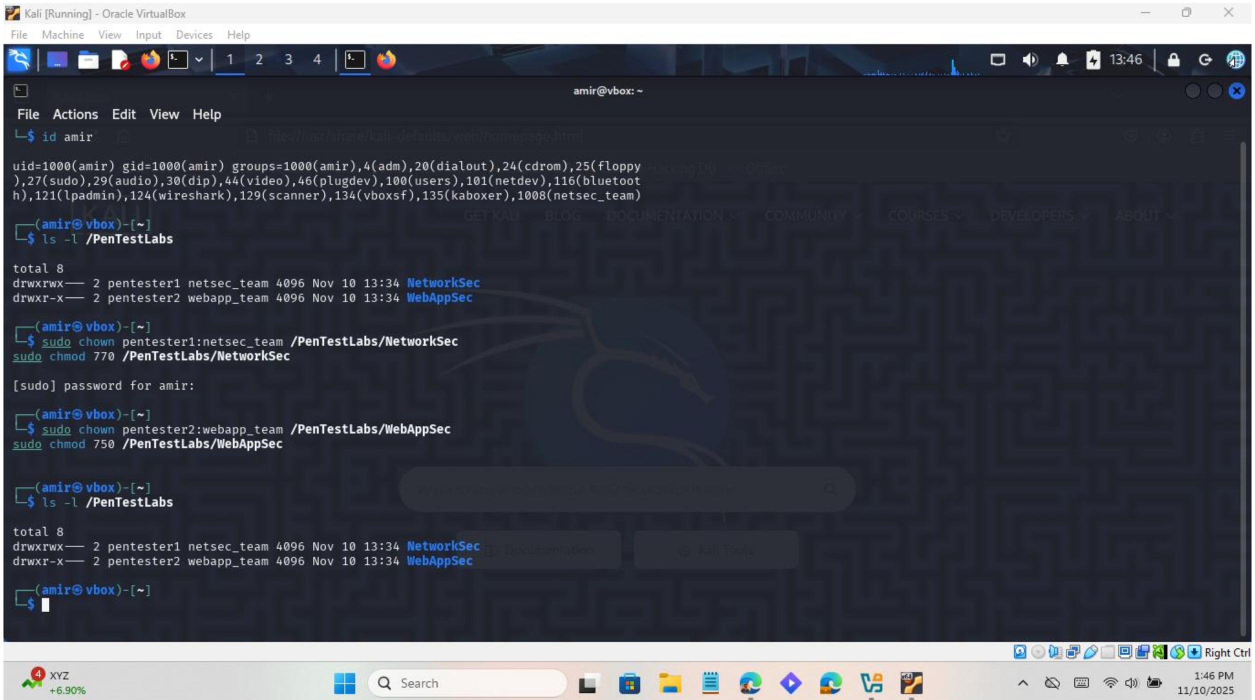
2.2. Set PenTestLabs permissions



2.3. Create users and groups

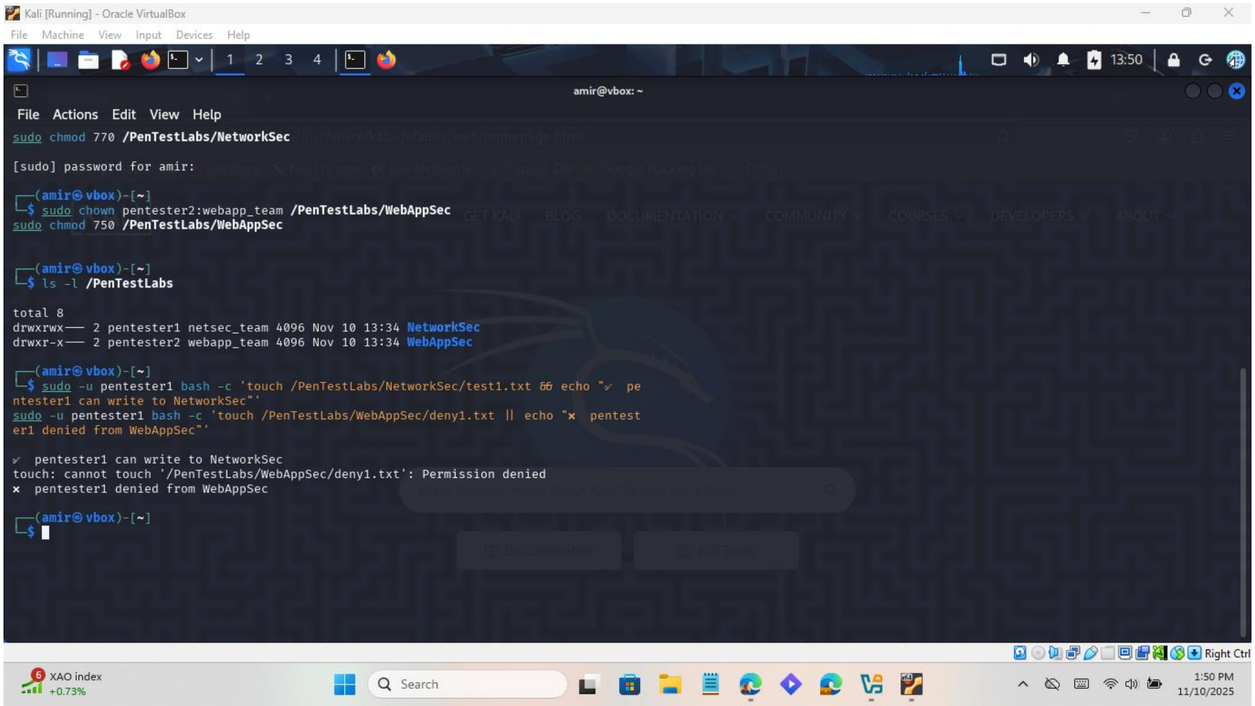


2.4. Set subdirectory permissions

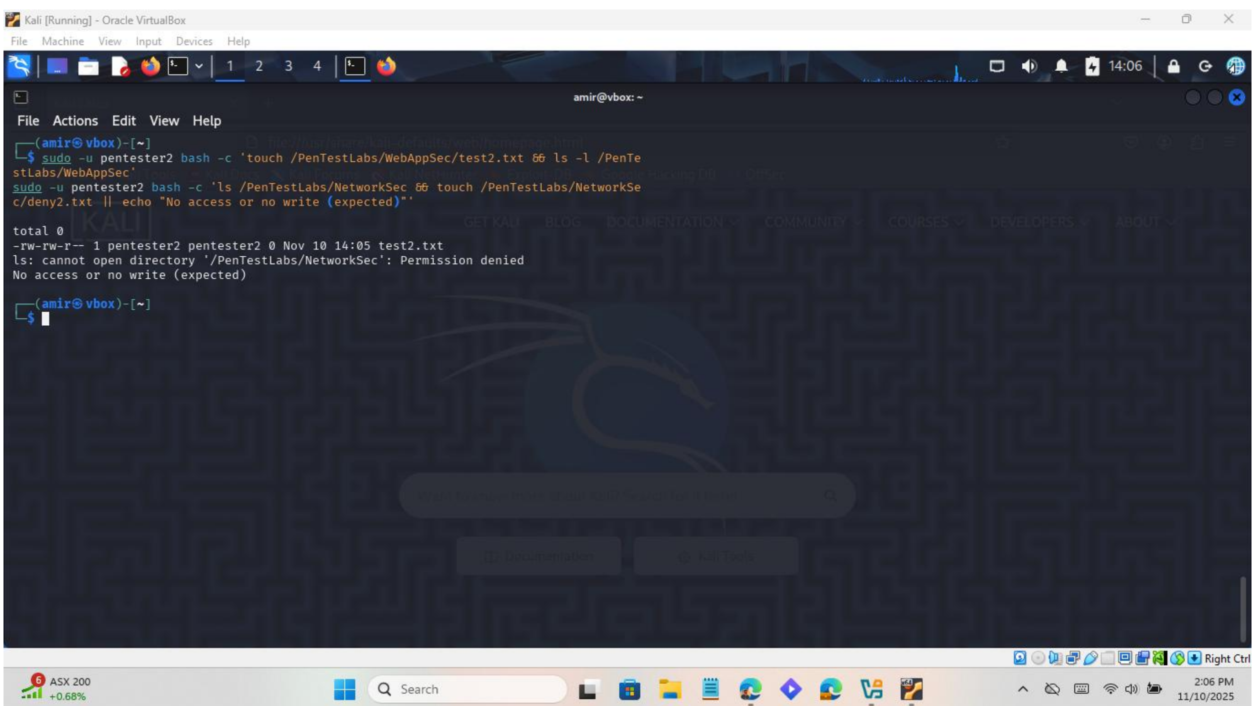


2.5. Verify with getfacl and user tests

As pentester1

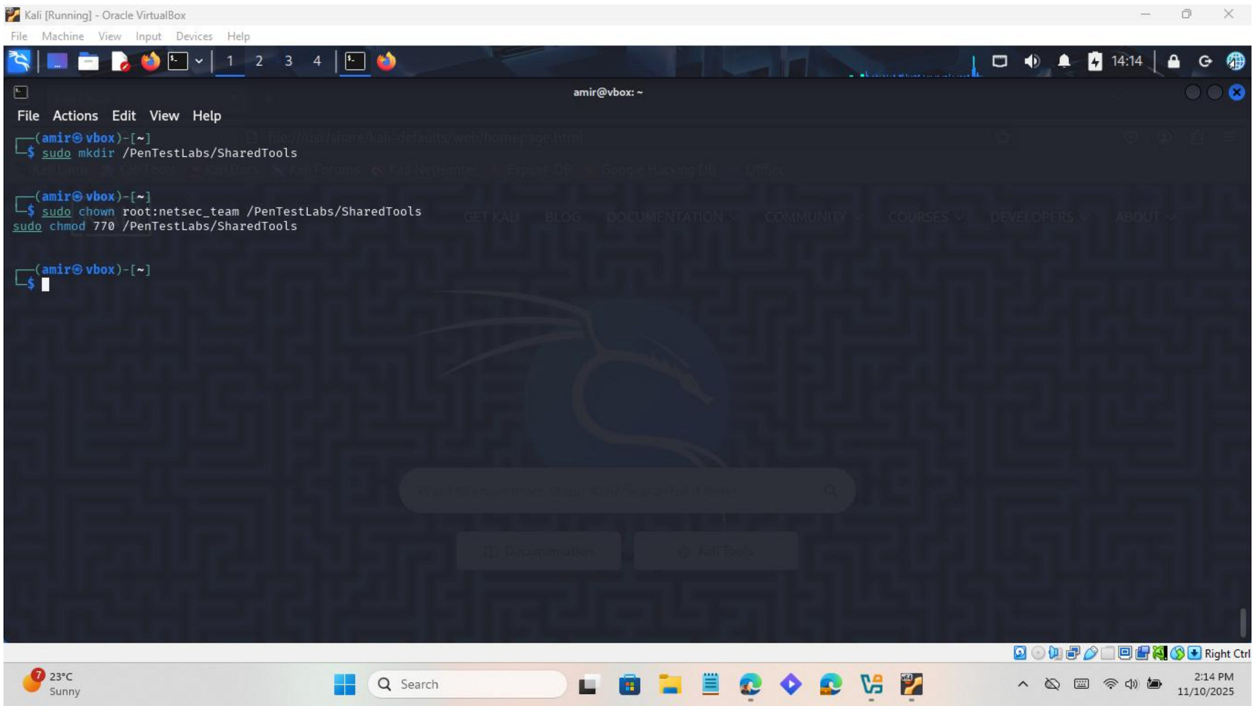


As pentester2

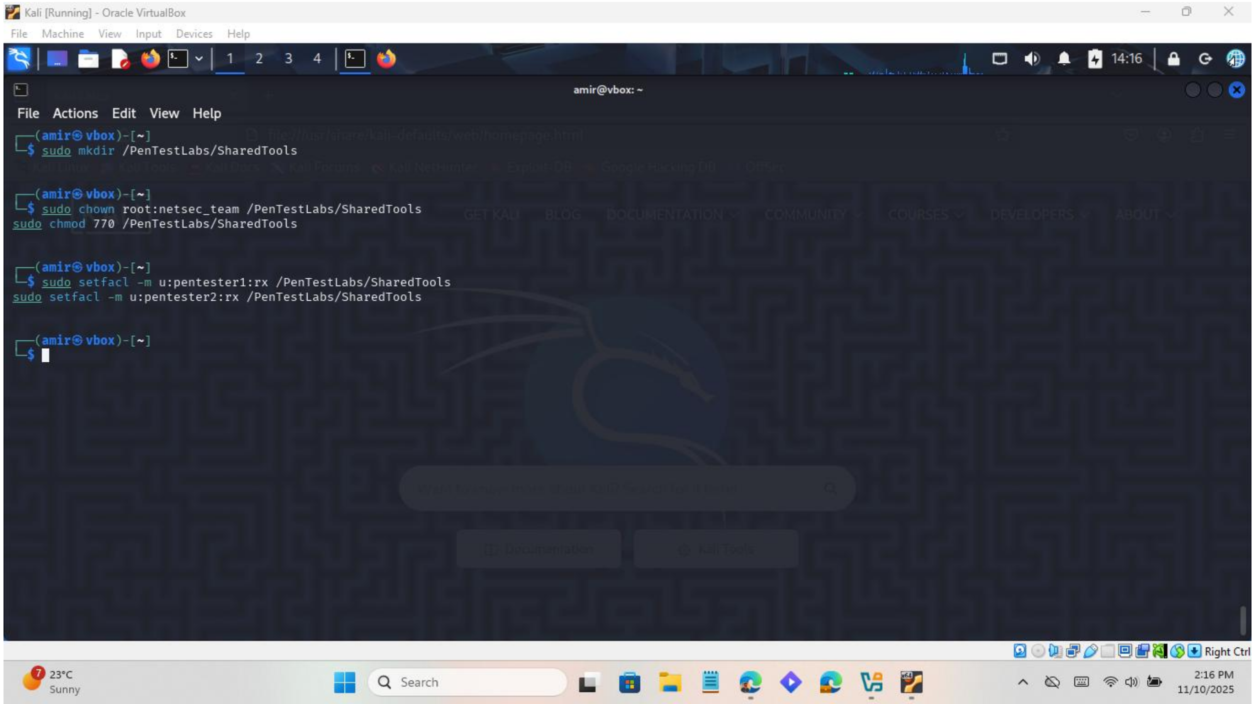


2.6. SharedTools: both can read/execute; only netsec_team can write

2.6.1. Create the directory and set base ownership and permissions

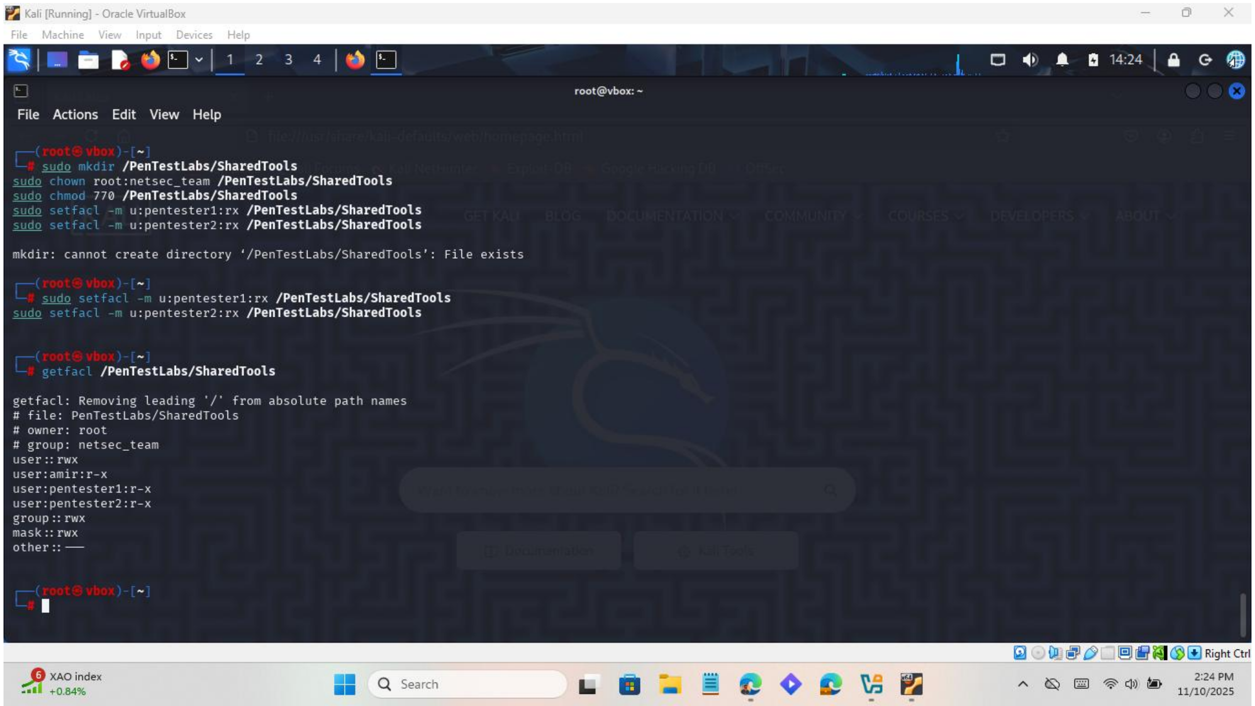


2.6.2. Apply ACLs for read/execute access



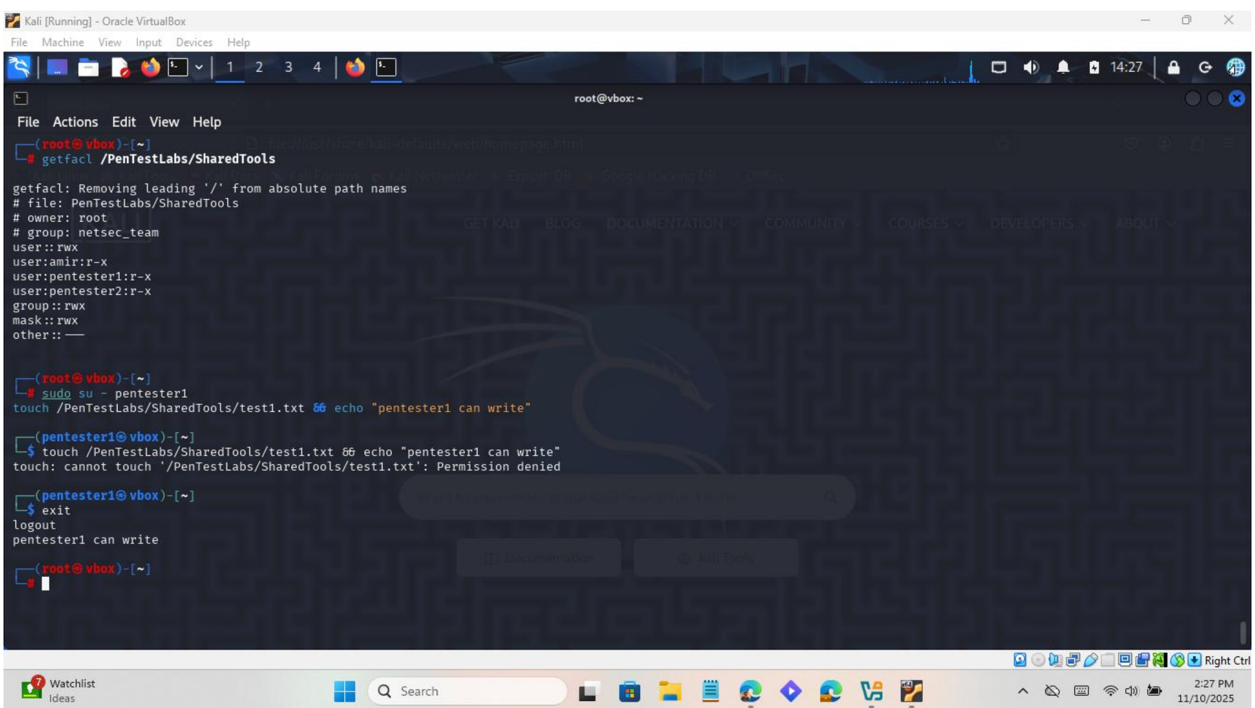
This allows both users to read and enter the directory but not write.

2.6.3. Verify ACLs

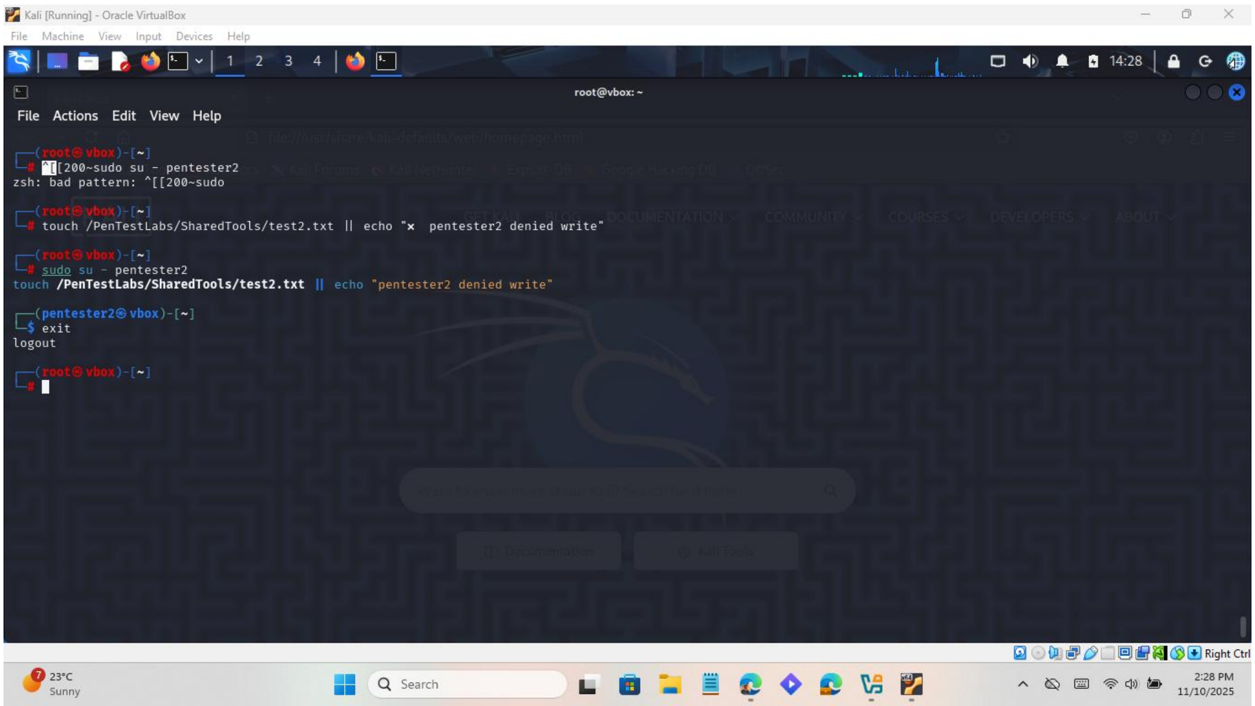


2.6.4. Access tests

<u>As pentester1 (netsec_team member)</u>

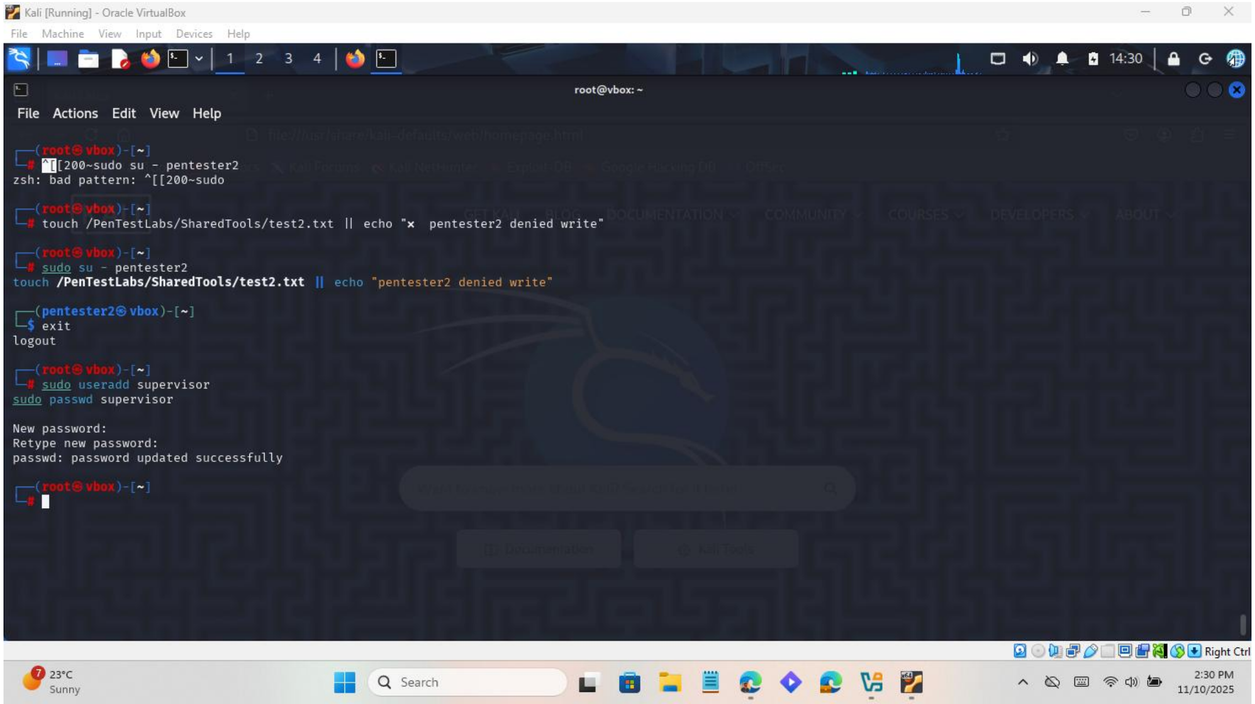


As pentester2 (not in netsec_team)

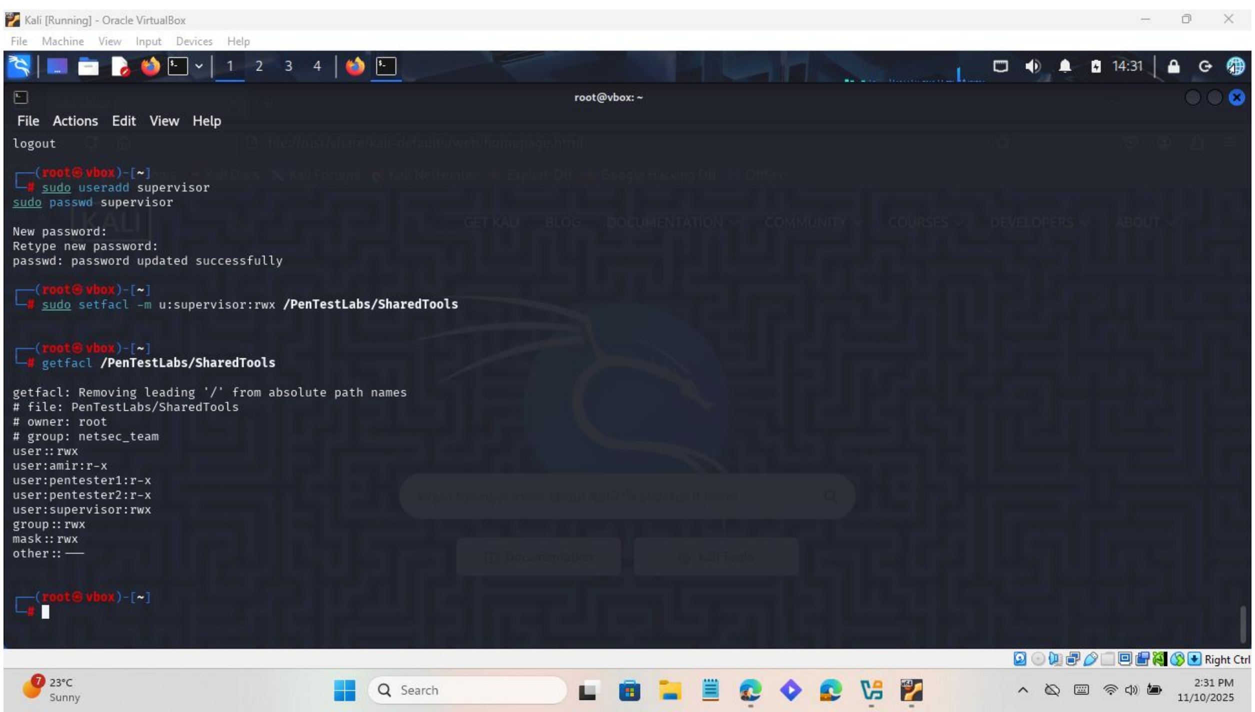


2.7. Supervisor read-only across all

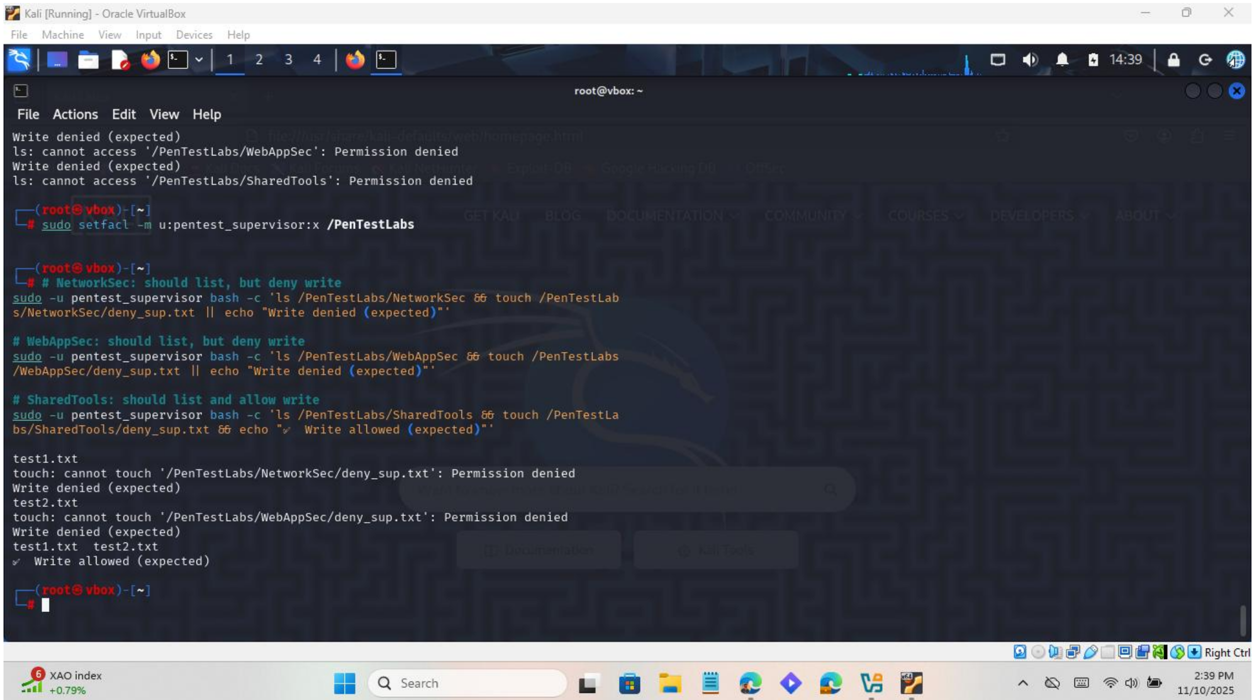
2.7.1. Create a supervisor user



2.7.2. Apply ACL for full access and verify ACL

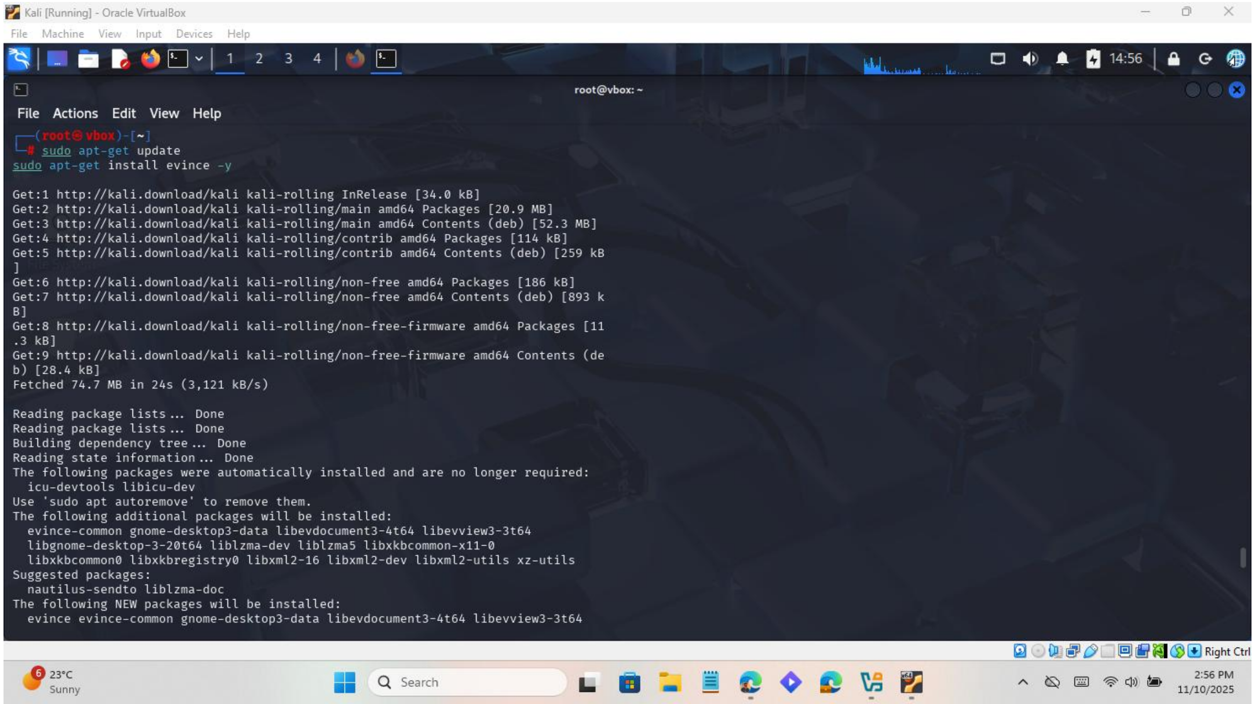


2.7.3. Test



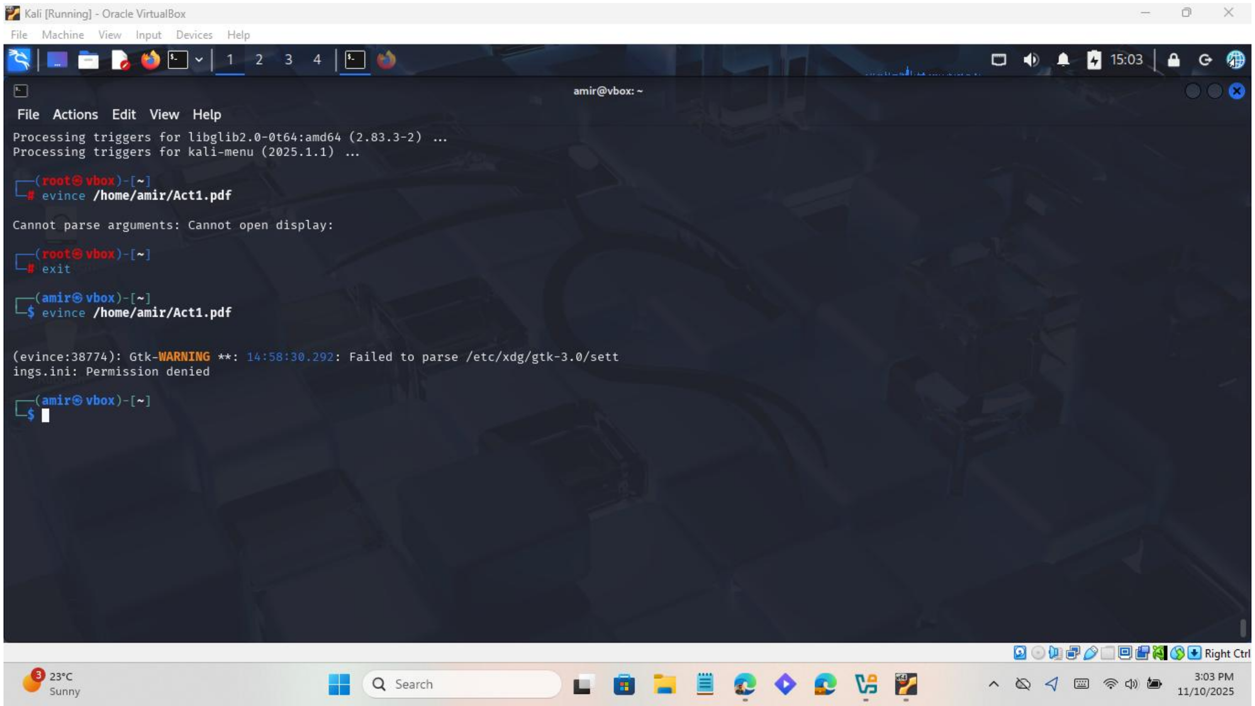
Q3. Password cracking

3.1. Install a PDF viewer



3.2. Act1.pdf: password profiling (social engineering)

3.2.1. Open Act1.pdf file in pdf viewer



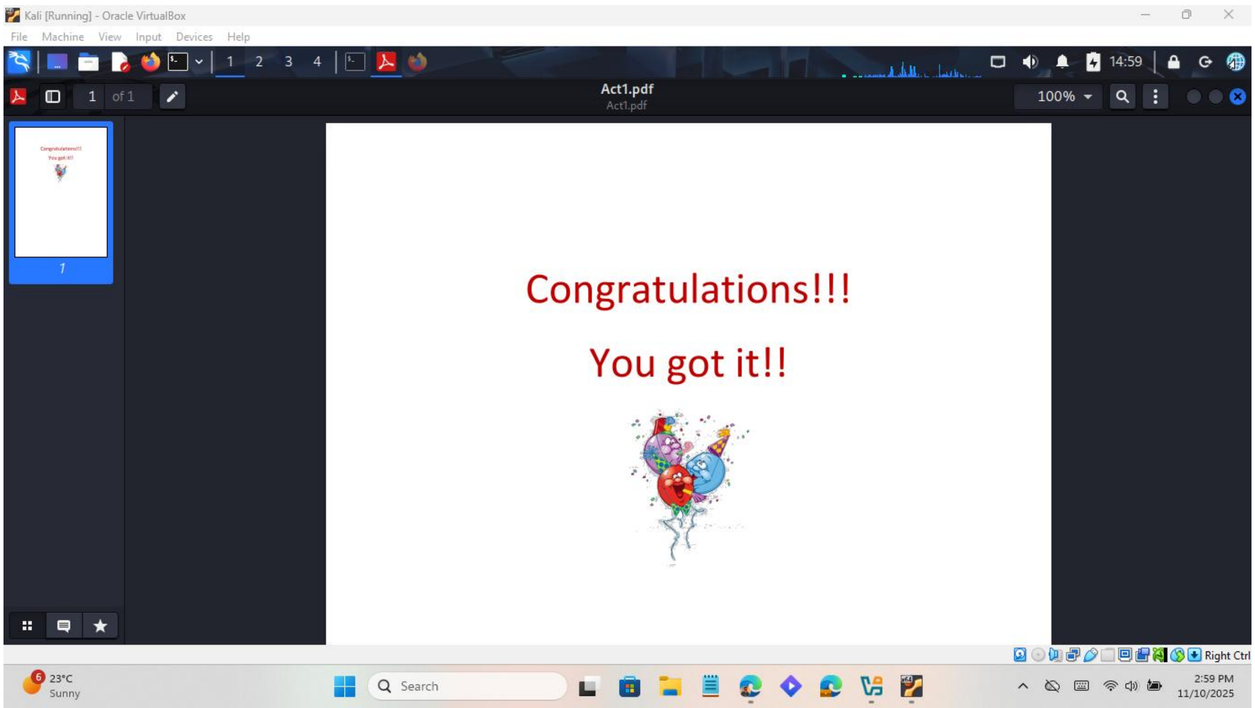
Based on the conversation:

- Jenny + favorite color “purple” + teacher/school cues. Try common human patterns at the PDF prompt:
- purple
- Purple
- jenny
- Jenny
- purplejenny
- jennypurple
- teacher
- english
- school

- purpleglasses

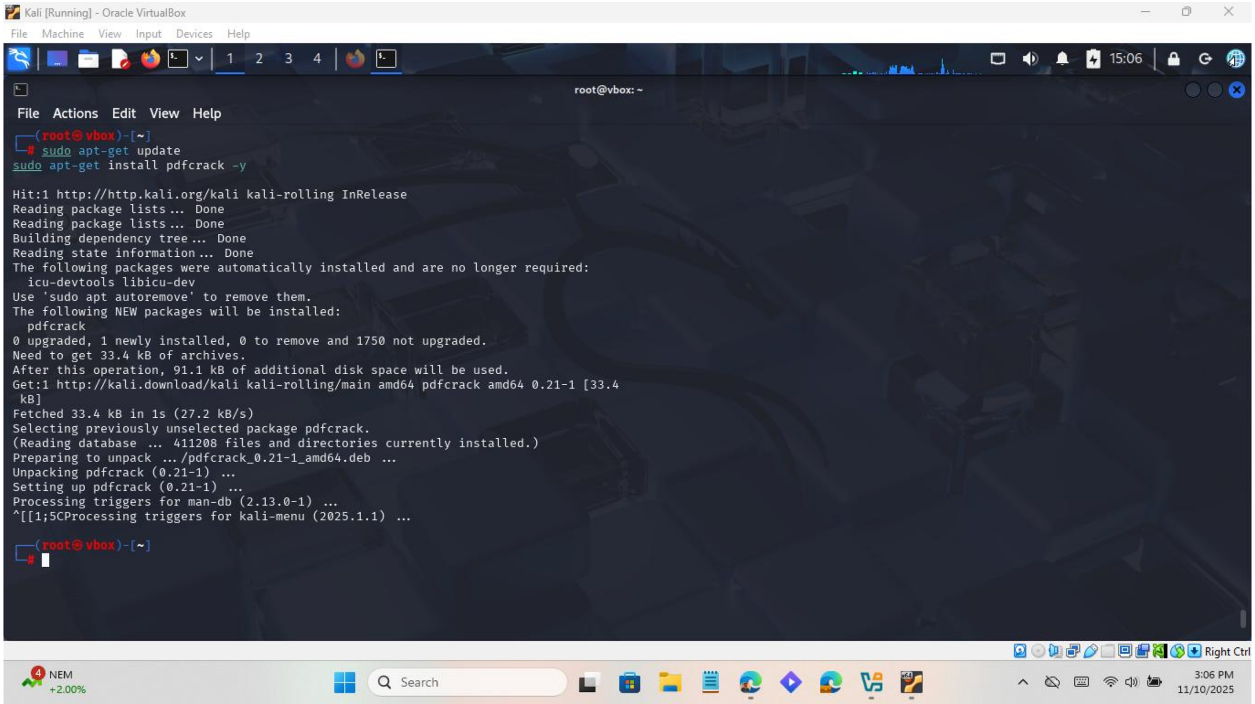
I tried all these password candidates to crack the password. So, the password for the Act1.pdf was **“purple”.** This password unlocked the file.

3.2.2. Unlocked Act1.pdf

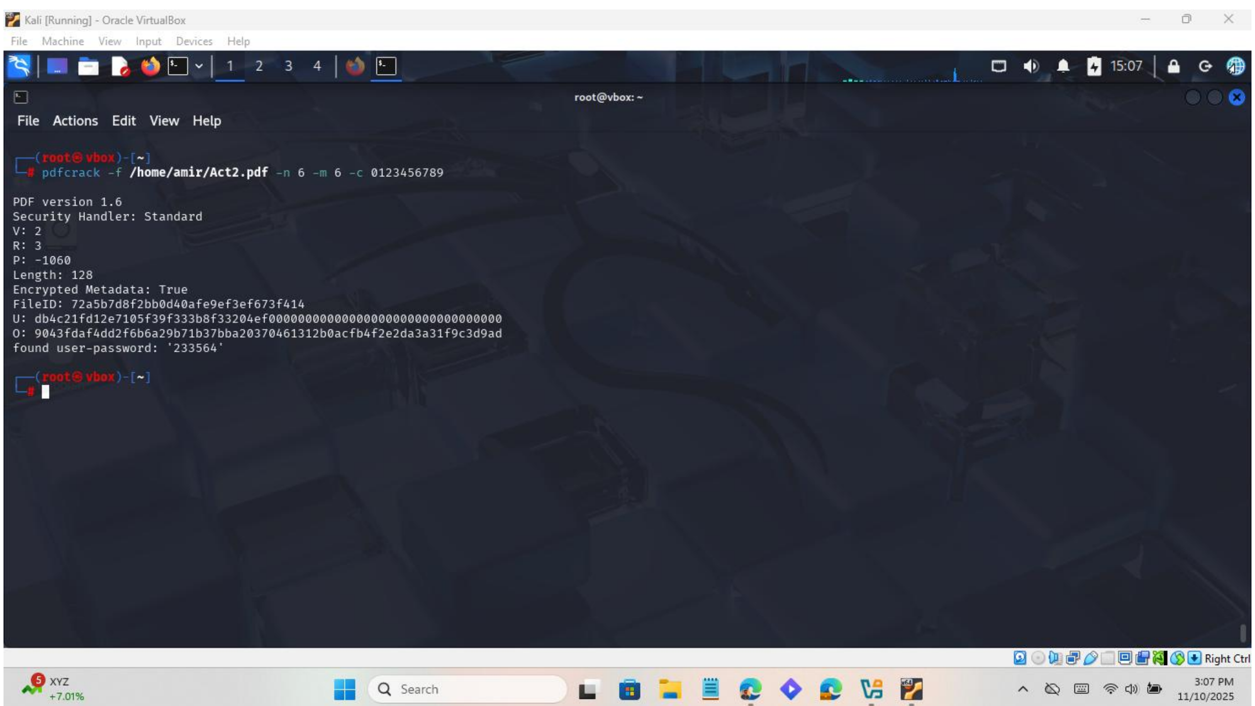


3.3. Act2.pdf: 6-digit numeric brute-force in Kali

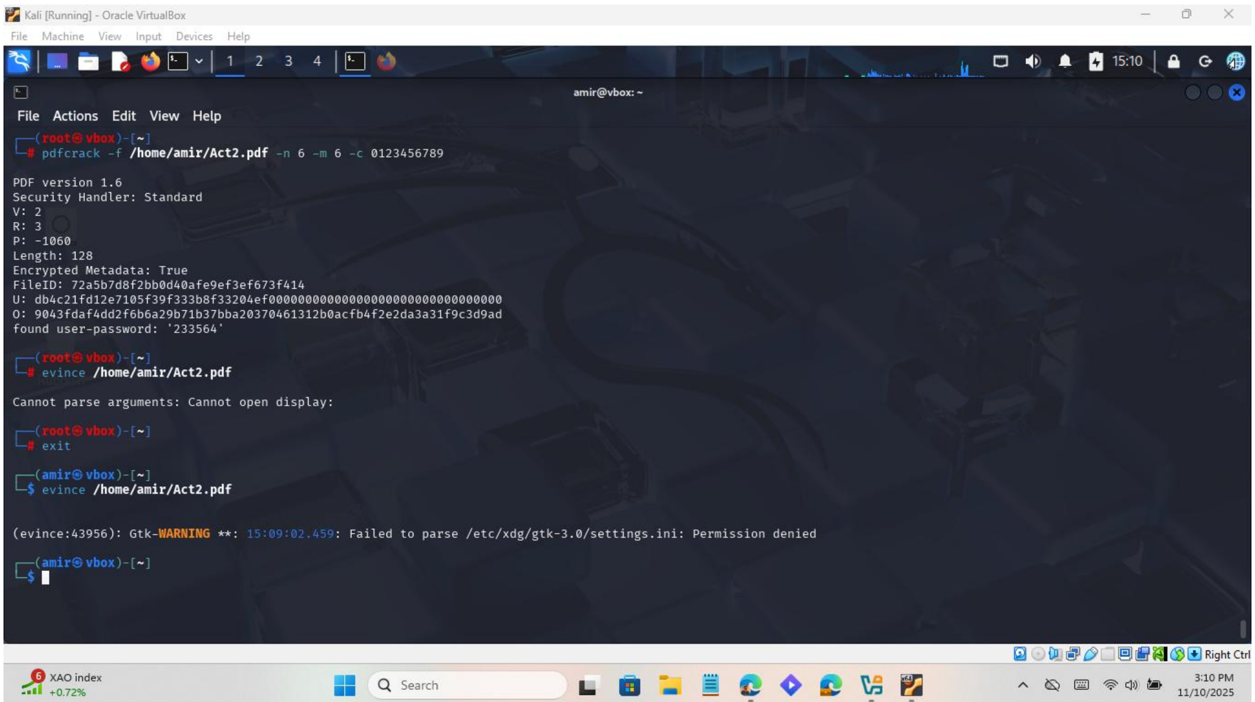
3.3.1. Install pdfcrack



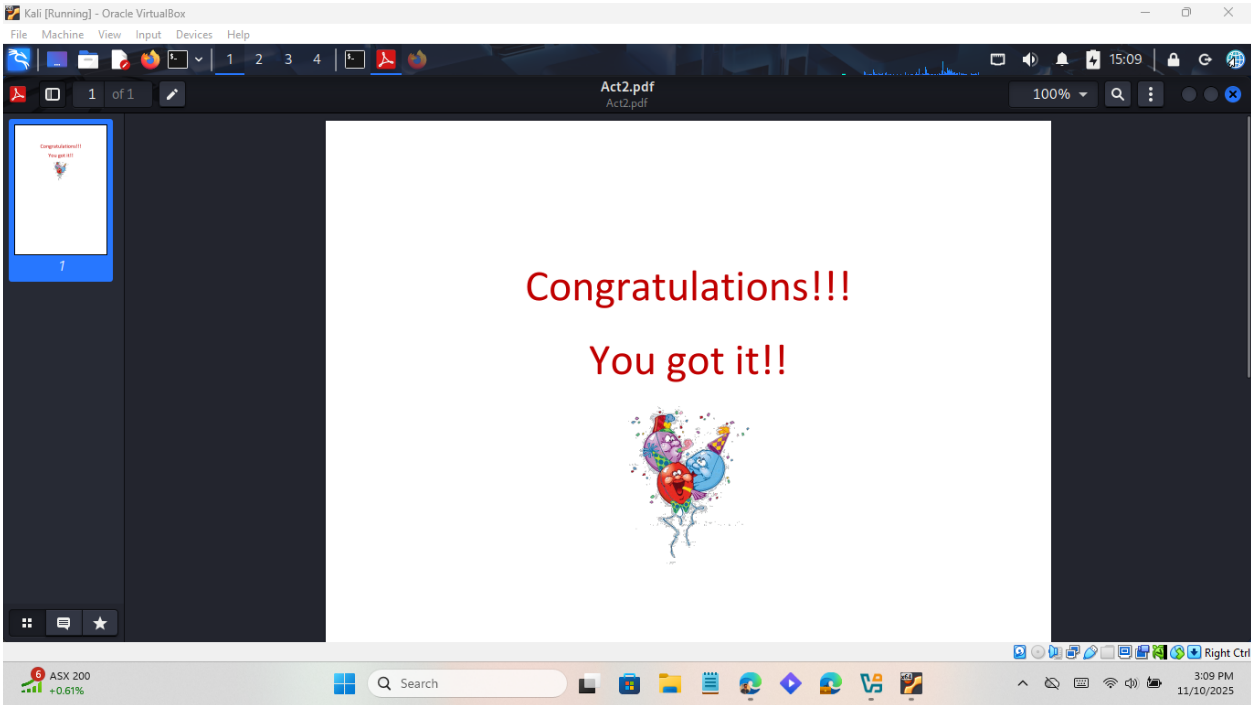
3.3.2. Run the brute-force attack



3.3.3. Open the PDF with the cracked password



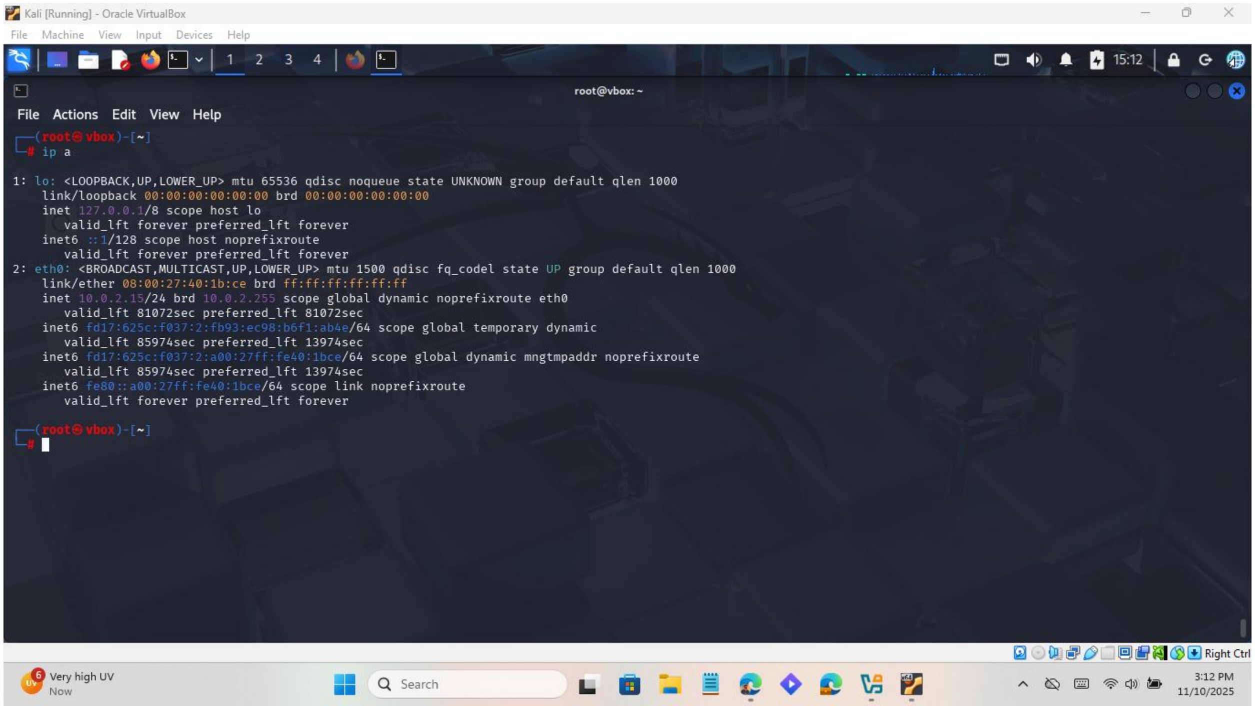
Result:



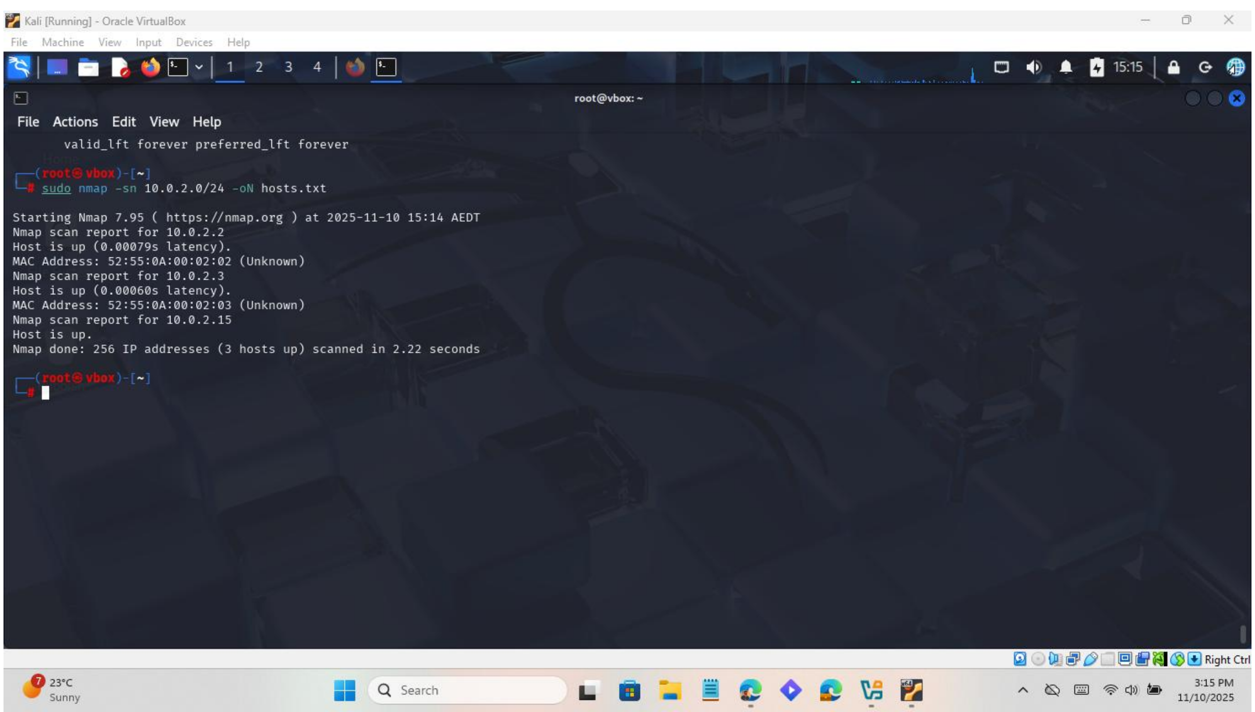
Q4. Nmap reconnaissance and NSE vulnerability assessment

4.1. Discover active hosts (ping sweep)

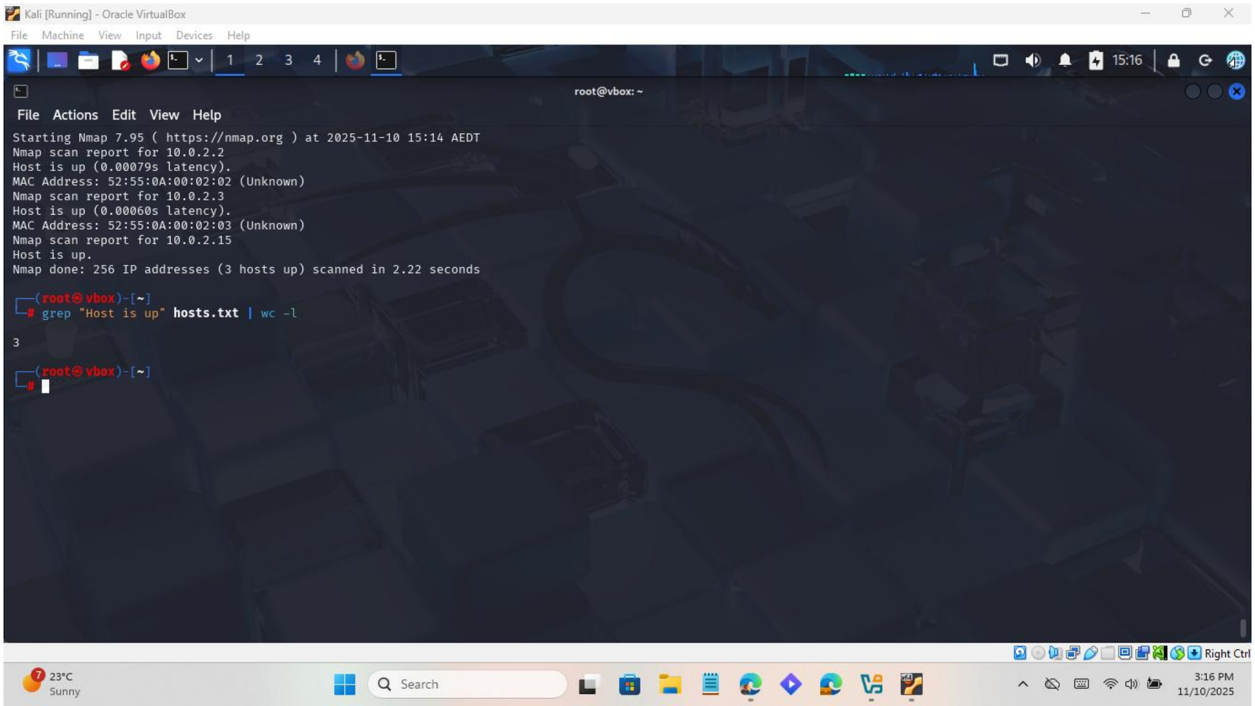
4.1.1. Identify subnet:



4.1.2. Run sweep

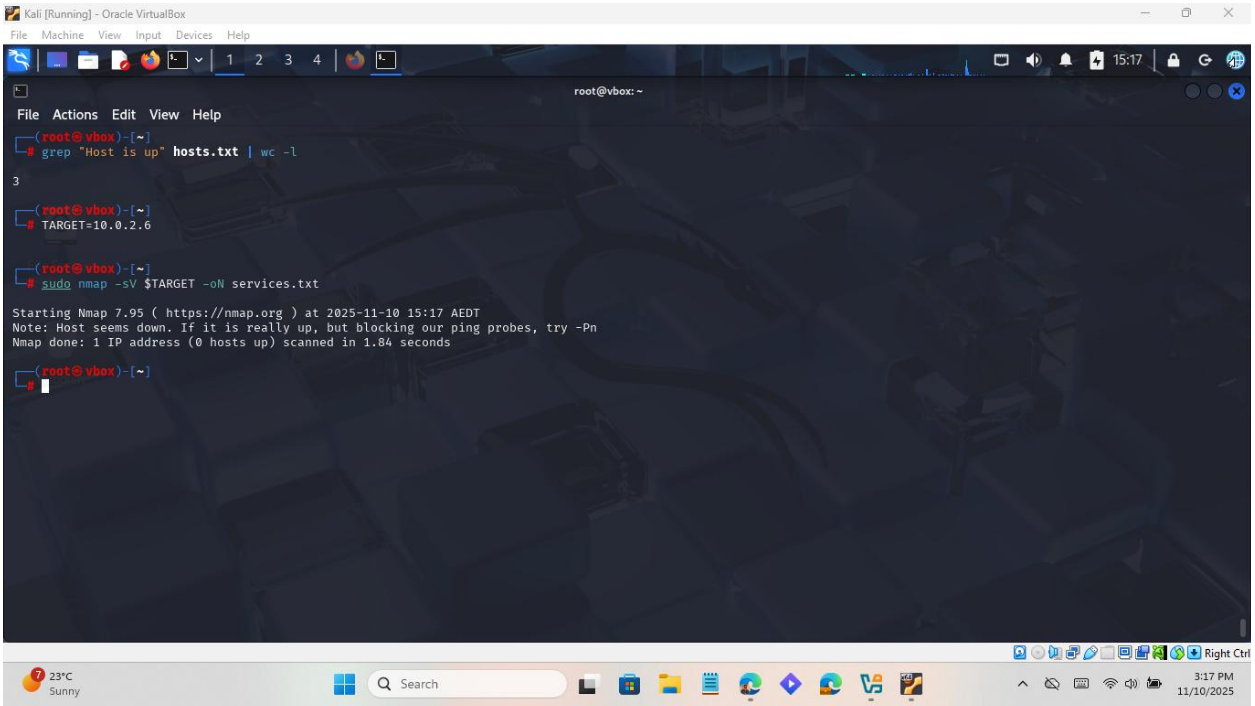


4.1.3. Count active hosts

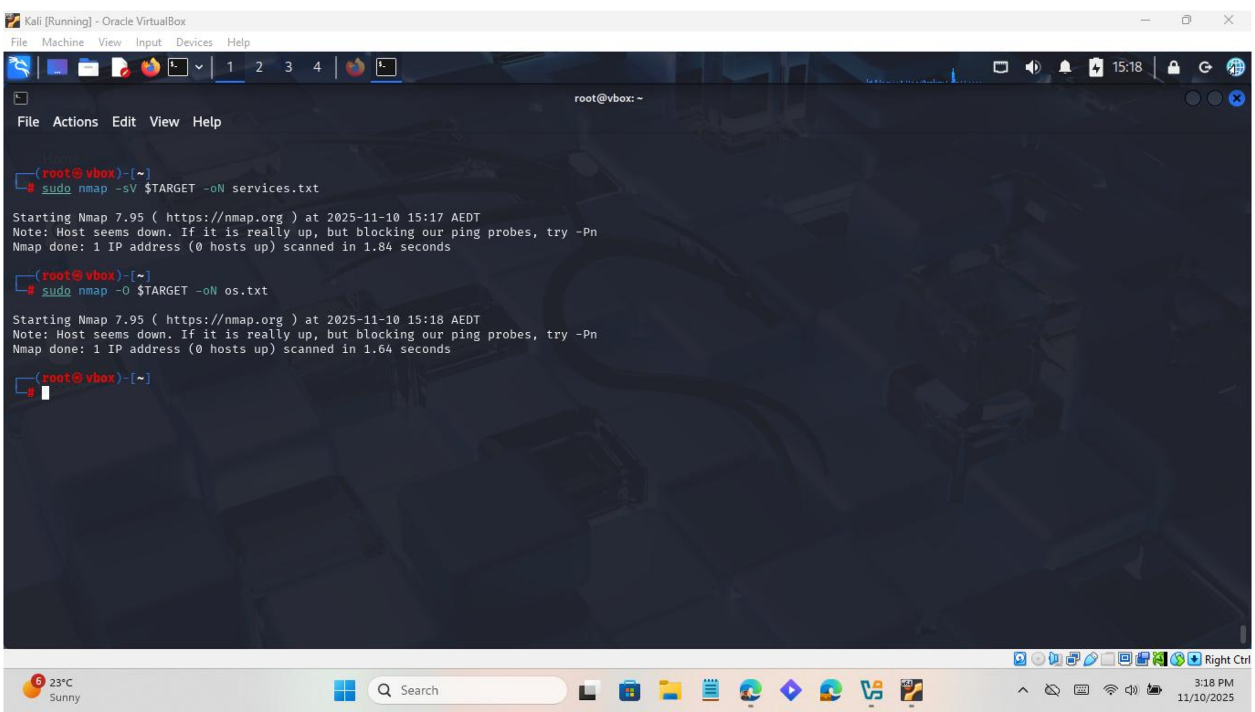


4.1.4. Service version detection on one host

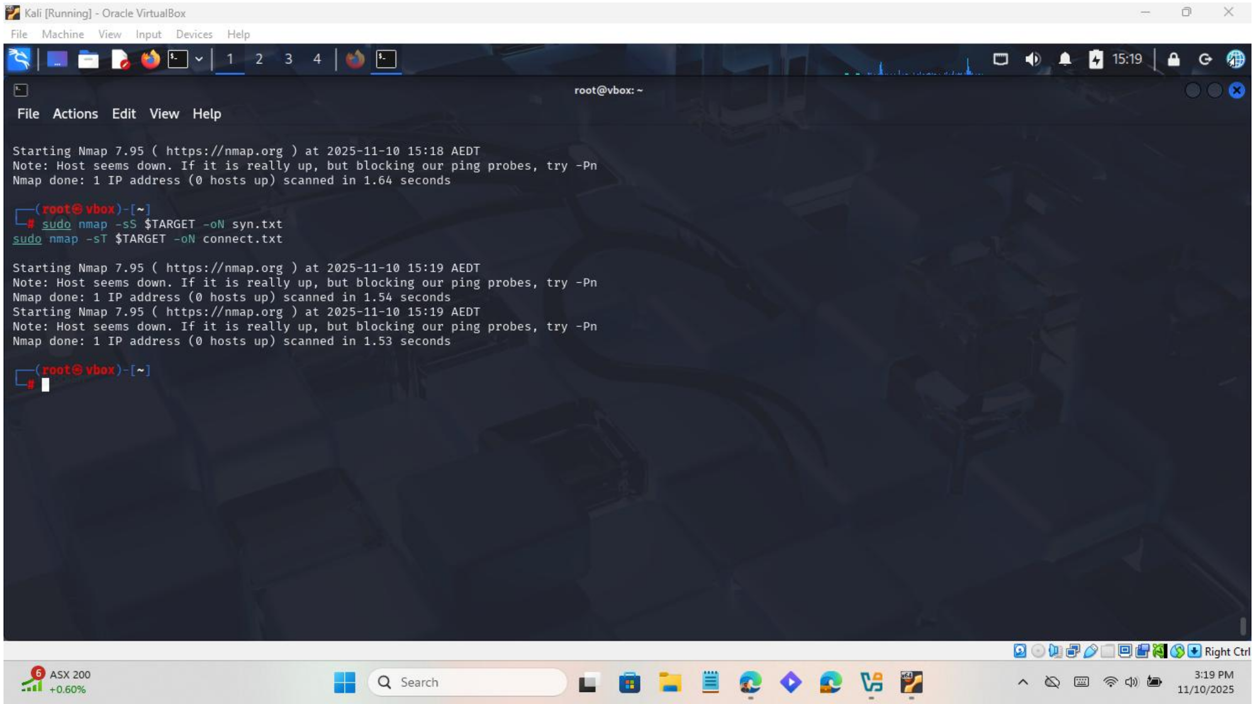
I chose one target IP from the sweep and run service scan



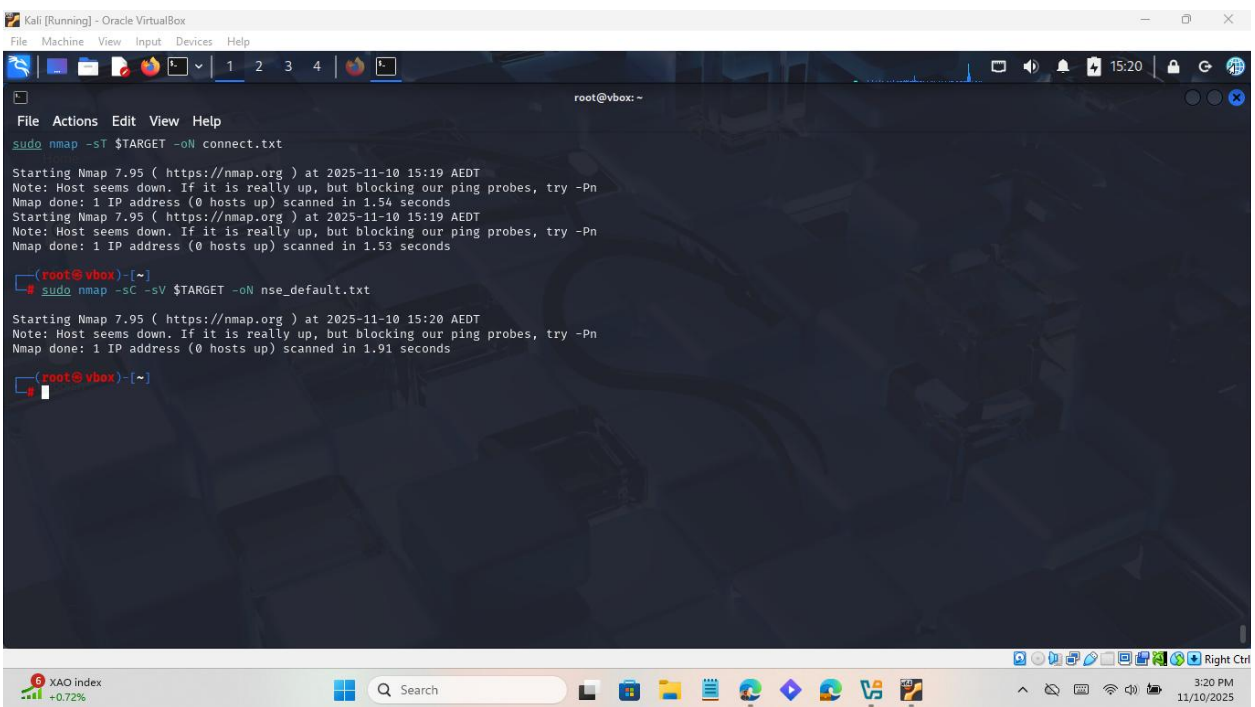
4.1.5. OS fingerprinting



4.1.6. SYN stealth vs TCP connect



4.1.7. NSE default scripts (vulnerability indications)



Reflection snippet to include

Reconnaissance helps defenders identify exposed services, outdated versions, and misconfigurations before attackers exploit them. Pairing recon with patching, firewall rules, and monitoring significantly reduces the attack surface.